Name : Adhip Halder

University Roll : 18871024030

Stream : MCA

Year : 2nd

Semester : 3rd

Subject Name : Software Engineering using UML

Subject Code : MCAN301

❑ About Software Engineering

➢ Definition

Software Engineering is a disciplined approach to the development, operation, and maintenance of software using engineering principles.

➢ Objectives

- Build reliable, scalable, and efficient software systems
- Improve productivity and reduce development time
- Ensure high-quality deliverables that meet customer needs

➢ Principles of Software Engineering

- Modularity : Breaking down into manageable parts
- Abstraction : Simplifying complex processes
- Reusability : Promoting reuse of components
- Scalability : Handling increased load efficiently
- Maintainability : Easy to update or fix

➢ Common Software Engineering Models:

- ✓ Waterfall Model
- ✓ Agile Model
- ✓ Spiral Model
- ✓ V-Model

❑ Phases in Software Engineering

➢ Requirement Gathering & Analysis

- Identify user needs, functional and non-functional requirements
- Techniques: Interviews, Questionnaires, Use Cases

➢ System Design

- Architecture design, data flow diagrams, database schema
- High-level and low-level design

➢ Implementation (Coding)

- Programming the design into working software
- Follows coding standards and language-specific conventions

➢ Testing

- Unit Testing, Integration Testing, System Testing, Acceptance Testing
- Ensures the system meets requirements

➢ Deployment

- Software is installed and made operational for users

➢ Maintenance

- Bug fixes, updates, and feature enhancements

❑ What is a Use Case Diagram?

➢ Definition

A Use Case Diagram models the interactions between users and the system to achieve specific goals.

➢ Why Use It?

- ✓ Provides a high-level functional view
- ✓ Easy to understand for both technical and non-technical stakeholders
- ✓ Helps identify system boundaries and required features

➢ Benefits

- Improves requirement clarity
- Aids in system documentation
- Useful in testing and validation

➢ Elements of a Use Case Diagram:

- **Actors:** External entities interacting with the system (e.g., User, Admin)
- **Use Cases:** Specific actions/functions the system performs
- **System Boundary:** Defines what is included in the system
- **Relationships:** Include, Extend, Generalization

❑ How to Draw a Use Case Diagram

Example System: Library Management System

- **Identify Actors**
  - Example: Librarian, Member (Student/Teacher), System Administrator
- **Define Use Cases**
  - Issue Book
  - Return Book
  - Borrow Book
  - Add New Book
  - Manage Memberships
- **Draw System Boundary**
  - A box labeled *"Library System"* containing all use cases
- **Connect Actors to Use Cases**
  - Librarian → Issue Book, Return Book, Add New Book
  - Member → Borrow Book, Return Book
  - Admin → Manage Memberships